應數二離散數學 2026 春, 第一次期中考

學號: ________________________________, 姓名: ________________________________

本次考試共有 9 頁 (包含封面)，有 13 題。如有缺頁或漏題，請立刻告知監考人員。

考試須知:

- 請在第一及最後一頁填上姓名學號。
- **不可**翻閱課本或筆記。
- 計算題請寫出計算過程，閱卷人員會視情況給予部份分數。
  沒有計算過程，就算回答正確答案也可能得到零分。
  答卷請清楚乾淨，儘可能標記或是框出最終答案。

高師大校訓：**誠敬宏遠**

誠，一生動念都是誠實端正的。 **敬**，就是對知識的認真尊重。
宏，開拓視界，恢宏心胸。 **遠**，任重致遠，不畏艱難。

請尊重自己也尊重其他同學，考試時請勿東張西望交頭接耳。

1. (10 points) Find the sum $\sum_{k=0}^{n} \binom{n}{k} r^k$ for all n.

   Answer: ______________________________.

2. (10 points) How many ways are there to arrange the letters in the word "VISITING" such that no two I's are consecutive?

   Answer: ______________________________.

3. (10 points) Determine the number of integral solutions of the equation

$$x + y + z + w = 20$$

that satisfy

$$1 \leq x \leq 5, \ \ -2 \leq y \leq 4, \ 3 \leq z \leq 8, \ 4 \leq w \leq 7.$$

Answer: ____.

4. (10 points) Construct a permutation of $\{1, 2, \ldots, 9\}$ whose inversion sequences are 6, 2, 5, 4, 1, 2, 0, 1, 0.

   The resulting permutation is: ______________________________.

5. (10 points) A committee of 6 people is to be chosen from 9 men and 7 women. (a) How many ways if there are no restrictions? (b) How many ways if the committee must contain at least 4 women?

   Answer: (a) ____________________, (b) ____________________.

6. (10 points) Find the position of the 3-subset $\{2, 4, 7\}$ of $\{1, 2, 3, 4, 5, 6, 7, 8\}$ in lexicographic order.

   Answer: ______________________________.

7. (10 points) Find the coefficient of x^5 in the expansion of $(2 - x + 3x^2)^8$.

   Answer: ______________________________.

8. (10 points) Find the number of integers between 1 and 1000 (inclusive) that are not divisible by 4, 5, or 6.

   Answer: ______________________________.

9. (10 points) Find the number of derangements of $\{1, 2, 3, 4, 5\}$ that start with the number 5.

   Answer: ______________________________.

10. (10 points) Given 150 integers $a_1, a_2, \ldots, a_{150}$, there exist integers r and s with $0 \leq r < s \leq 150$ such that $a_{r+1} + a_{r+2} + \cdots + a_s$ is divisible by 150. Prove this statement.

11. (10 points) Show that among any $n+1$ integers chosen from $\{1, 2, \ldots, 2n\}$, there are always two numbers such that one divides the other.

12. (10 points) Use combinatorial reasoning to prove the identity:

$$\sum_{k=0}^{n} \binom{r+k}{k} = \binom{r+n+1}{n}$$

13. (10 points) How many permutations of the multiset $\{1, 1, 2, 2, 3, 3, 4, 4\}$ are there such that no two identical digits are adjacent? (沒有任何兩個相同數字相鄰)

Answer: ____.

學號: ____________, 姓名: ____________, **以下由閱卷人員填寫**

Question:	1	2	3	4	5	6	7	8	9	10	11	12	13	Total
Points:	10	10	10	10	10	10	10	10	10	10	10	10	10	130
Score:														